

Assignment 1

For University Dataset Questions.

1. Find the names of all the students whose total credits are greater than 100 .

Query:- SELECT name FROM student WHERE tot_cred>100;

```
mysql> SELECT name FROM student WHERE tot_cred>100;
+-----+
| name |
+-----+
| Zhang |
| Chavez |
| Tanaka |
+-----+
3 rows in set (0.00 sec)
```

mysql> █

0 sudo 1:sudo* Assignment1

2. Find the course id and grades of all courses taken by any student named 'Tanaka'.

Query:- SELECT t.course_id, t.grade FROM takes as t join student as s on s.ID=t.ID WHERE s.name='Tanaka';

```
mysql> SELECT t.course_id, t.grade FROM takes as t join student as s on s.ID=t.ID WHERE s.name=
'Tanaka';
+-----+-----+
| course_id | grade |
+-----+-----+
| BIO-101 | A |
| BIO-301 | NULL |
+-----+-----+
2 rows in set (0.01 sec)
```

mysql> █

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3. Find the courses which are offered in both 'Fall' and 'Spring' semester (not necessarily in the same year).

Query:- SELECT c.course_id, c.title FROM course c JOIN section s ON c.course_id = s.course_id WHERE s.semester IN ('Fall', 'Spring') GROUP BY c.course_id, c.title HAVING COUNT(DISTINCT s.semester) = 2;

```
mysql> SELECT c.course_id, c.title FROM course c JOIN section s ON c.course_id
-> = s.course_id WHERE s.semester IN ('Fall', 'Spring') GROUP BY c.course_id, c.title HAVIN
G COUNT(DISTINCT s.semester) = 2;
+-----+
| course_id | title                               |
+-----+
| CS-101    | Intro. to Computer Science |
+-----+
1 row in set (0.00 sec)

mysql>
cc 0 ^ sudo 1:sudo* Assignment1
```

4. Find the names of all the instructors from Comp. Sci. department.

Query:- SELECT name FROM instructor WHERE dept_name='Comp. Sci.';

```
mysql> SELECT name FROM instructor WHERE dept_name='Comp. Sci.';
+-----+
| name      |
+-----+
| Srinivasan |
| Katz       |
| Brandt     |
+-----+
3 rows in set (0.00 sec)

mysql>
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```

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5. Find the course id and titles of all courses taught by an instructor named 'Srinivasan'

Query:- SELECT c.course_id,c.title FROM course as c JOIN teaches as t On c.course_id=t.course_id JOIN instructor as i ON i.ID=t.ID WHERE i.name='Srinivasan';

```
mysql> SELECT c.course_id,c.title FROM course as c JOIN teaches as t On c.course_id=t.course_id
JOIN instructor as i ON i.ID=t.ID WHERE i.name='Srinivasan';
+-----+-----+
| course_id | title                                |
+-----+-----+
| CS-101    | Intro. to Computer Science         |
| CS-315    | Robotics                           |
| CS-347    | Database System Concepts           |
+-----+-----+
3 rows in set (0.00 sec)

mysql>
```

6. Find the names of instructors who have taught at least one course in Spring 2009.

Query:- SELECT DISTINCT i.name FROM instructor as i JOIN teaches as t ON i.ID=t.ID WHERE t.semester='Spring' AND t.year=2009;

```
mysql> SELECT DISTINCT i.name FROM instructor as i JOIN teaches as t ON i.ID=t.ID WHERE t.semester='Spring' AND t.year=2009;
+-----+
| name |
+-----+
| Brandt |
| Kim   |
+-----+
2 rows in set (0.01 sec)

mysql>
```

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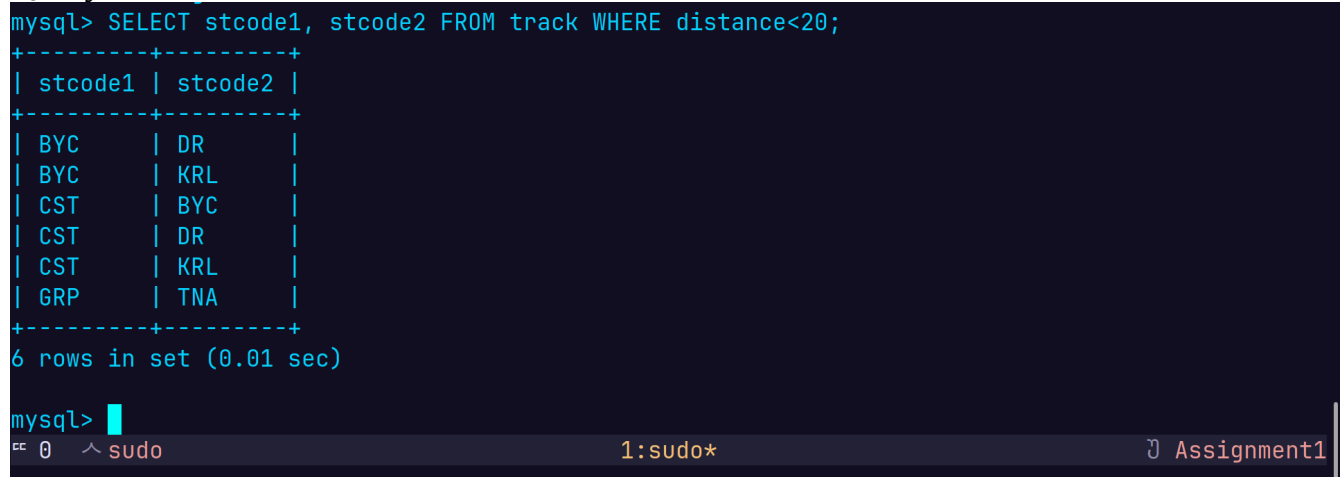
For Railway Dataset,

1. Find pairs of stations (station codes) that have a track (direct connection) with distance less than 20Kms between them.

Query:- `SELECT stcode1, stcode2 FROM track WHERE distance<20;`

```
mysql> SELECT stcode1, stcode2 FROM track WHERE distance<20;
+-----+-----+
| stcode1 | stcode2 |
+-----+-----+
| BYC     | DR      |
| BYC     | KRL     |
| CST     | BYC     |
| CST     | DR      |
| CST     | KRL     |
| GRP     | TNA     |
+-----+-----+
6 rows in set (0.01 sec)

mysql>
```

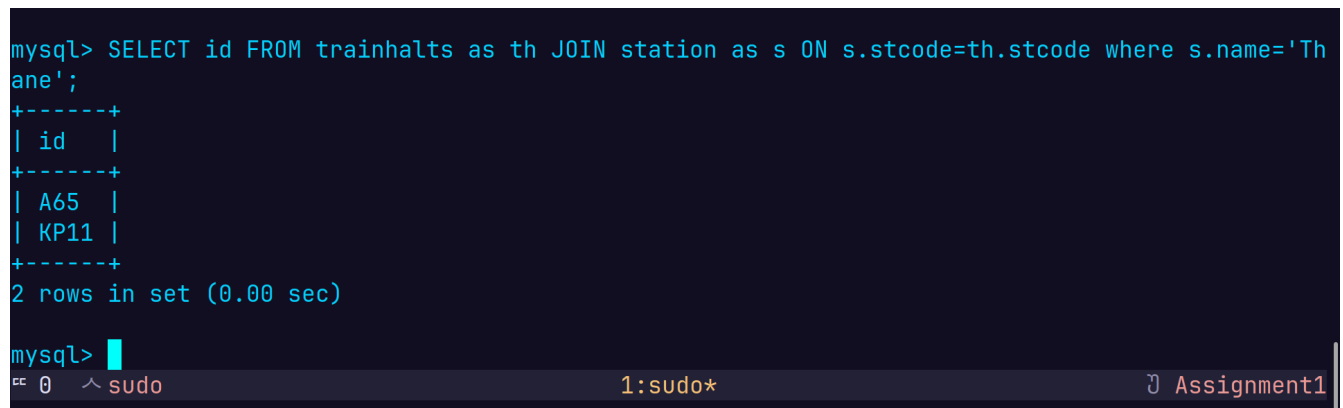


2. Find the IDs of all the trains which have a stop at THANE

Query:- `SELECT id FROM trainhalts as th JOIN station as s ON s.stcode=th.stcode where s.name='Thane';`

```
mysql> SELECT id FROM trainhalts as th JOIN station as s ON s.stcode=th.stcode where s.name='Thane';
+-----+
| id   |
+-----+
| A65  |
| KP11 |
+-----+
2 rows in set (0.00 sec)

mysql>
```



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3. Find the names of all trains that start at MUMBAI.

Query:- SELECT t.name FROM train as t JOIN trainhalts th ON t.id=th.id JOIN station as s ON th.stcode=s.stcode WHERE th.seqno=0 AND s.name='MUMBAI';

```
mysql> SELECT t.name FROM train as t JOIN trainhalts th ON t.id=th.id JOIN station as s ON th.stcode=s.stcode WHERE th.seqno=0 AND s.name='MUMBAI';
+-----+
| name      |
+-----+
| CST-AMR_LOCAL |
| CST-KYN      |
+-----+
2 rows in set (0.00 sec)

mysql>
```

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4. List all the stations in order of visit by the train 'CST-AMR_LOCAL'.

Query:- SELECT st.name FROM trainhalts AS th JOIN station AS st ON st.stcode=th.stcode WHERE id = (SELECT t.id FROM train AS t WHERE t.name = 'CST-AMR_LOCAL') ORDER BY seqno;

```
mysql> SELECT st.name FROM trainhalts AS th JOIN station AS st ON st.stcode=th.stcode WHERE id = (SELECT t.id FROM train AS t WHERE t.name = 'CST-AMR_LOCAL' -> ) ORDER BY seqno;
+-----+
| name      |
+-----+
| MUMBAI     |
| BYCULLA    |
| DADAR      |
| KURLA      |
| GHATKOPAR  |
| THANE      |
| DOMBIVALI  |
| KALYAN     |
| AMBARNATH  |
+-----+
9 rows in set (0.00 sec)

mysql>
```

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